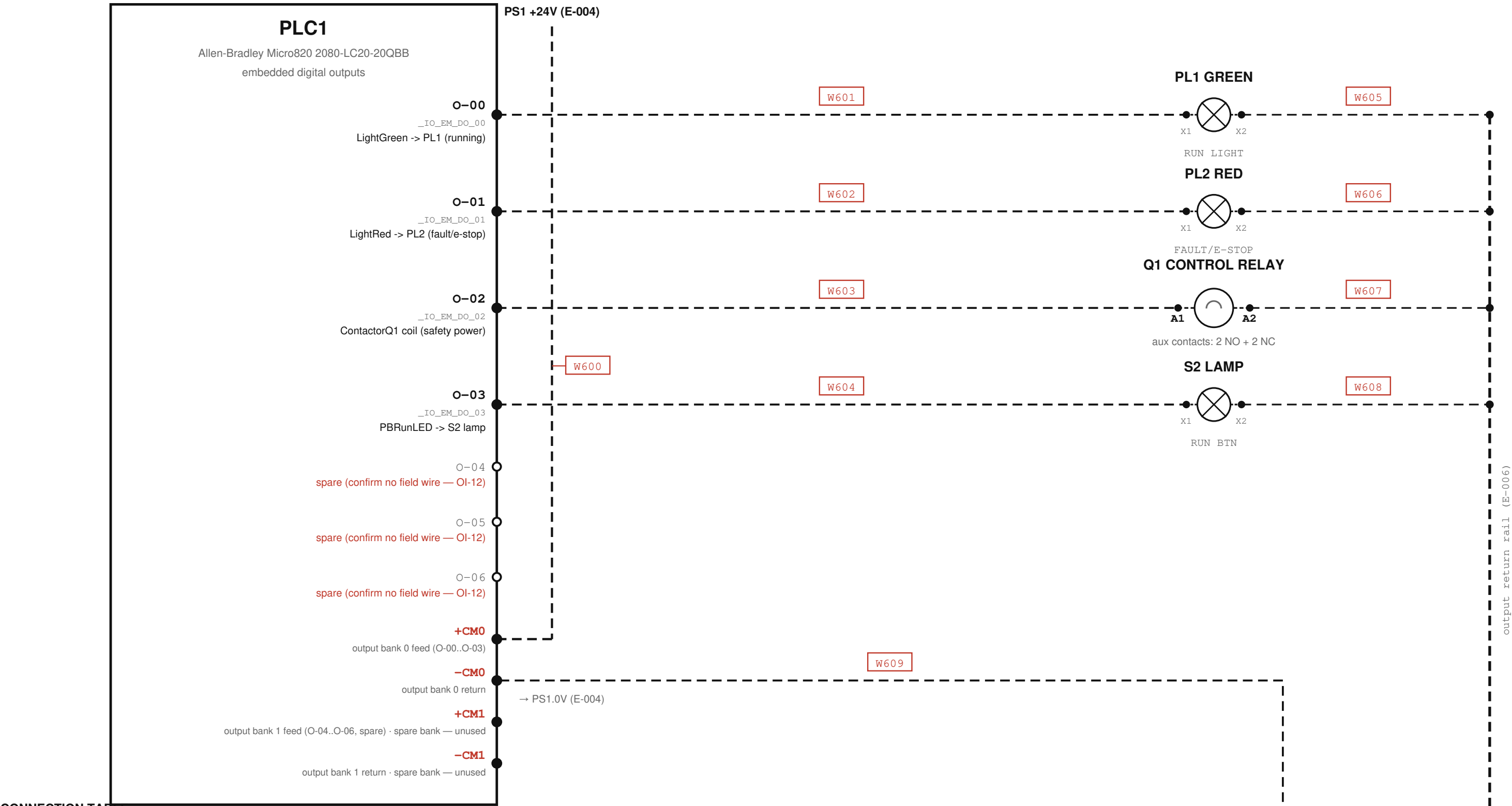


E-006 PLC OUTPUTS

Conv_Simple / CV-101 (garage conveyor) — O-00..O-06 · pilots, Q1/MLC control-relay coil, button lamp · GS10 control = HYBRID Modbus + hardwired I/O (E-007; map OI-22)



Hardwired fallback documented in MIRA-WI-001 v3 §3 Fig 1 (DO_03..DO_07 → GS10 DI1..DI5 + DCM) — NOT ACTIVE and NOT drawn: P02.0x at factory default per the 2026-05-20 parameter export; DO_07 does not exist on the 2080-LC20-20QBB (O-00..O-06); DO_03 collides with PBRunLED in the live I/O map. Bench presence of any such wiring = FIELD VERIFY (OI-18).

SAFETY

- Monitored outputs are NOT a safety function.
- E-stop must remove power per NFPA 79.
- Q1/MLC is a CONTROL RELAY (24 VDC coil, driven by O-02) whose 2 NO contacts (13-14, 43-44) switch the drive's 230 V 1φ supply (V6 correction, E-003) — de-energizing O-02 DOES remove Q1's switched supply to the GS10. Q1 is NOT an NFPA-79/EN-60204-1 safety-rated disconnect (small control-relay contacts, not a lockable safety contactor); the e-stop (S0) → O-02 relationship is FIELD VERIFY (OI-26). Do not treat this circuit as the LOTO isolation point — always LOTO at CB1 or upstream.
- O-02 is the sole MLC/Q1 drive-enable coil output — do not reuse it for any other load (MIRA_PLC_WorkInstruction_v3.pdf §3: "DO_02 is already the MLC drive-enable coil and must never be reused").

NOTES

- Q1/MLC = Schneider TeSys CA3KN22BD control relay (2 NO + 2 NC, 24 VDC coil); coil A1(+)/A2(-) driven by O-02 (verified from photo wire_3). NO contacts 13-14/43-44 switch the VFD's 230 V 1φ supply (see E-003); 21-22/31-32 NC unused.
- GS10 control (full-res photo): run/dir/speed = Modbus — top FWD/REV/DI connector is EMPTY, RJ45 landed. Bottom analog/output row (+10V/ACM/AI/AO1/DO1/DOC) is wired — likely speed reference + feedback. Map = OI-22.
- NO contacts 13-14 & 43-44 switch the 230 V 1φ VFD supply (see E-003); coil A1/A2 from O-02. NC 21-22/31-32 unused.
- GS10 run/dir/freq = Modbus — see E-007 (P00.20=1, P00.21=2 per 2026-05-20 parameter export)
- No as-built: O-00/O-01/O-03 from CCW v4.0; O-02 corroborated by live Prog_init v2.1.
- Output bank technology + common feed = FIELD VERIFY (OI-09): WI-001 p.4 says relay dry contacts; ±CM polarity naming suggests DC-fed banks — conflicting evidence, resolve per 2080-IN009 + meter.
- GS10 manual warns against cycling a power-circuit switching device for normal run/stop — emergency situations only (GS10_UM.txt L1754-1757, L1811-1813). This applies to Q1/MLC (energized by O-02, switches the drive's 230 V 1φ supply — see E-003): O-02/Q1 stays energized during normal operation; run/stop is commanded via Modbus (E-007), never by cycling Q1.
- S2 lamp (this sheet, O-03/PBRunLED) and S2's run/rearm pushbutton contact (E-005, I-04) are two elements of the SAME physical device — same cross-reference symmetry as Q1/MLC's coil (this sheet) ↔ power contacts (E-003).

SOURCES:

CCW_VARIABLES_v4.0.txt
Prog_init_ConvSimple_v2.1.st
MIRA_PLC_WorkInstruction_v3.pdf
GS10_actual_parameters_5.20.26.xlsx
GS10_UM.txt
photos review/photos/wire_1..4.jpg (2026-07-11, actual devices)
full-res photos review/photos/"_full.jpg + crops (2026-07-11);
review/PHOTO_EVIDENCE_V5.md

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- Wxxx proposed wire number (red = unverified)

CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W600	PS1.+24V	PLC1.+CM0	output bank 0 feed	power_24vdc	field_verify	
W601	PLC1.O-00	PL1.X1	LightGreen (run light)	control_output	field_verify	
W602	PLC1.O-01	PL2.X1	LightRed (fault/e-stop light)	control_output	field_verify	
W603	PLC1.O-02	Q1.A1	ContactorQ1 coil (safety power)	control_output	field_verify	
W604	PLC1.O-03	S2.X1	PBRunLED (run button lamp)	control_output	field_verify	
W605	PL1.X2	output return rail (E-006)	run light return	control_return	field_verify	
W606	PL2.X2	output return rail (E-006)	fault light return	control_return	field_verify	
W607	Q1.A2	output return rail (E-006)	contactor coil return	control_return	field_verify	
W608	S2.X2	output return rail (E-006)	PB lamp return	control_return	field_verify	
W609	PLC1.-CM0	PS1.0V	output bank 0 return	power_0v	field_verify	

PLC OUTPUTS

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM

output map CCW v4.0 + live Prog_init v2.1; O-02 do-not-reuse (MIRA_PLC_WorkInstruction_v3.pdf §3) — see SAFETY

SHEET

E-006

REV A

2026-07-11

6 of 9